Predictive Nephrotoxicity Screening in Small Molecules

A Multi-Model Cheminformatics Framework with Tree-SHAP Explainability

- **Under the Guidance of: Prof. Rajnish Kumar**
- Department of Chemical Engineering & Technology **Indian Institute of Technology (Banaras Hindu University), Varanasi**

- **Presentation Date: 7** July 2026

Notebook Colab : [https://github.com/sakshihulke11-oss/KidneyTox-QSAR/blob/main/Nephrotoxicity_smallcompounds%20(1).ipynb]

Why Nephrotoxicity Deserves an Early Computational Filter

Proximal Tubule Epithelium

Concentrates and reabsorbs filtered xenobiotics — the primary site of drug-induced kidney injury.

Late detection = high clinical and financial cost

Dose-limiting toxicity

Nephrotoxicity frequently caps how much of a drug can safely be given — directly constraining therapeutic dosing.

Detected late in development

Renal injury signals often surface only in Phase II/III trials, or post-market — after major capital is already committed.

A computational triage layer helps

In-silico pre-screening filters high-risk compounds before they enter costly wet-lab and clinical pipelines.

Five Objectives, One End-to-End Pipeline

1. **Engineer a unified feature space**
   Convert raw SMILES strings into a common tabular numerical representation.

2. **Benchmark three model families**
   Random Forest and XGBoost ensembles vs. a deep neural network, on identical data.

3. **Select the best performer**
   Use standardized diagnostics — not accuracy alone — to pick a production model.

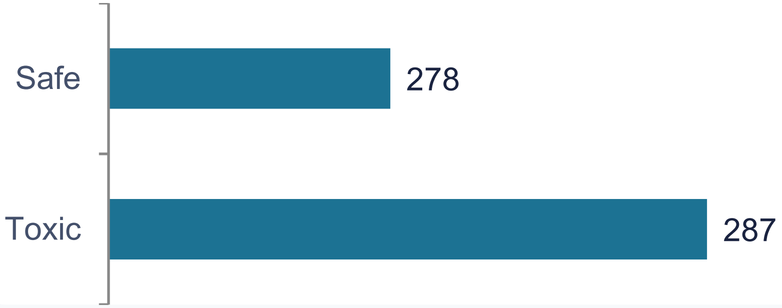
4. **Attribute predictions to structure**
   Apply Tree-SHAP to trace toxicity scores back to specific molecular features.

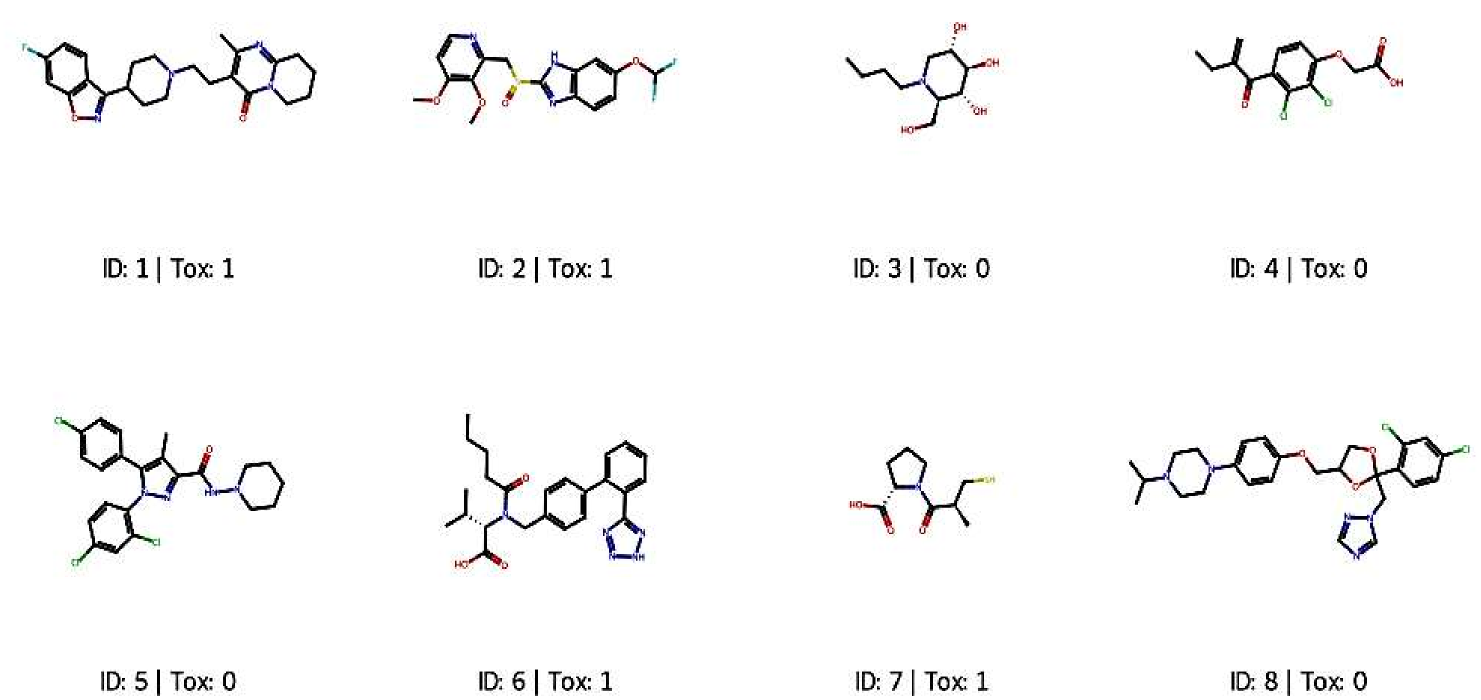
5. **Deploy a live inference engine**
   Package the pipeline into a callable function for screening unseen compounds.

565 Compounds, Near-Balanced, Structure-Labeled

565
Total Compounds

Class Distribution





First 8 compounds rendered directly from SMILES · RDKit (ID | Toxicity label)

Total Dataset Space: 565 Small Molecules.
Split Balance: Train Set = 452 compounds
Test Set = 113 compounds.

Note: class split is near-balanced (50.8% / 49.2%) — not severely skewed.

Dataset source: Amin, Kar & Piotto (2026). "KidneyTox_v1.0 enables explainable AI prediction of nephrotoxicity in small molecules." Scientific Reports 16:5099.

Two Chemical Languages, One Feature Matrix

Physicochemical Descriptors

Full RDKit descriptor suite

- LogP (lipophilicity)
- Molecular Weight
- Topological Polar Surface Area
- H-Bond Donors / Acceptors
- Ring counts & topological indices

Structural Fingerprint

Morgan circular fingerprint

- 2048-bit binary vector
- Radius = 2 (ECFP4-equivalent)
- Encodes local atomic neighborhoods
- Captures substructural motifs descriptors miss

↓ Concatenated ↓

Missingness Filter

< 20% NaN tolerance

→

Zero-Variance Filter

Removes uninformative columns

→

Final Feature Matrix

2175 columns / compound

Three Architectures, One Common Feature Space

Random Forest

Bagged decision tree ensemble

- n_estimators: 300 / 500
- max_depth: 10 / 15 / None
- min_samples_leaf: 1 / 2

XGBoost

Gradient-boosted tree ensemble

- n_estimators: 300 / 500
- learning_rate: 0.05 / 0.1
- max_depth: 4 / 6 / 8

Deep Neural Network

3 hidden layers, dropout-regularized

- Dense(256, ReLU) → Dropout(0.3)
- Dense(128, ReLU) → Dropout(0.3)
- Dense(64, ReLU) → Dropout(0.2) → Sigmoid

Three Firewalls Against Data Leakage

TRAINING PARTITION

Mutual Information filter

Fit exclusively on training data — selects top 300 of 2175 features

↓

SMOTE oversampling

Applied only to the training partition (k_neighbors = 5)

↓

5-fold GridSearchCV

Hyperparameter tuning performed strictly inside the training partition

TEST PARTITION

Never resampled.

Never used in feature selection.

Touched exactly once — at final evaluation.

Four Diagnostics, Because Accuracy Alone Isn't Enough

Accuracy

Overall correct-classification rate across both classes.

ROC-AUC

Discrimination ability across every possible probability threshold.

PR-AUC (Avg. Precision)

Precision-recall tradeoff — threshold-independent, emphasizes the toxic class.

F1-Score

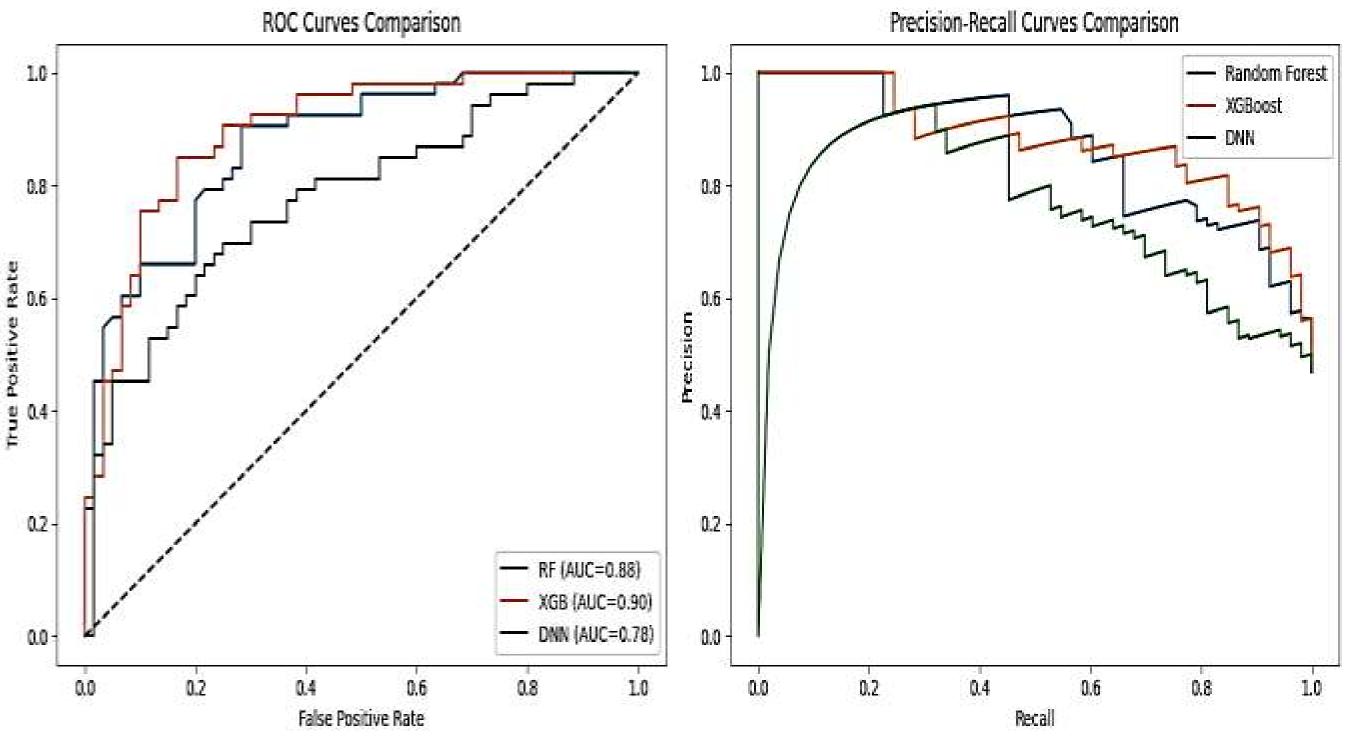
Harmonic balance between precision and recall.

08 · RESULTS

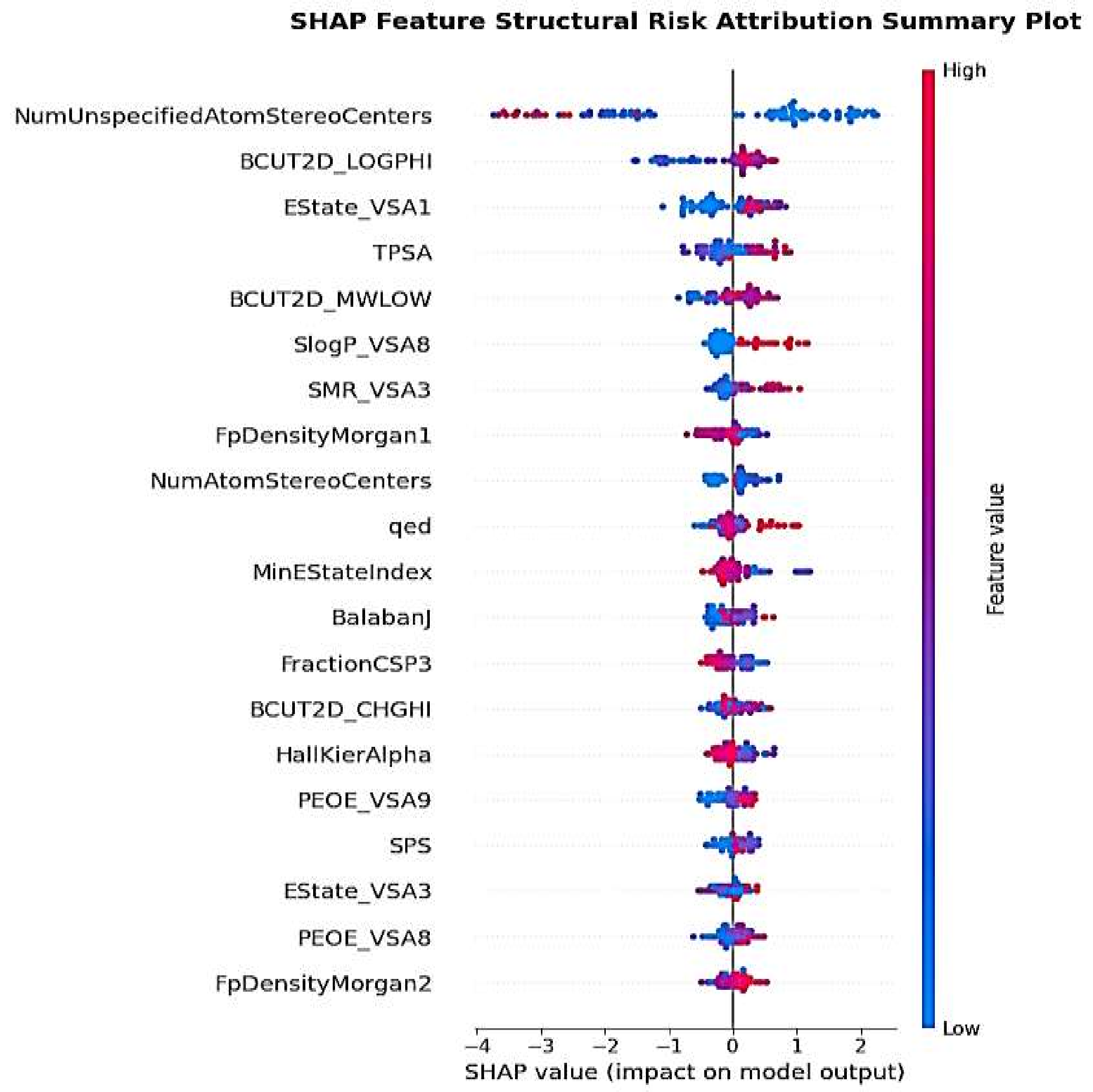
XGBoost Leads on Every Single Metric

Model	Accuracy	ROC-AUC	PR-AUC	F1-Score
Random Forest	0.788	0.876	0.870	0.774
XGBoost	0.814	0.898	0.880	0.811
DNN	0.699	0.776	0.752	0.685

- **5-fold Stratified CV confirms this isn't a lucky split: 0.821 ± 0.014 ROC-AUC across folds.**
- **DNN trails XGBoost by ~12 ROC-AUC points — expected given only ~450 training rows against a 300-feature input.**



Opening the Black Box with Tree-SHAP



- Tree-SHAP explainer applied directly to XGBoost — the tournament winner
- Beeswarm ranks 20 features by mean absolute Shapley value
- Point color = feature magnitude; position = direction of toxicity push
- Top driver: NumUnspecifiedAtomStereoCenters — unresolved stereochemistry pushes risk sharply in both directions
- Also prominent: BCUT2D_LOGPHI, EState_VSA1, TPSA — polarity and charge-distribution descriptors

Live Screening on an Unseen Compound

Aspirin

```
Target Compound Query (Aspirin):
CC(=O)OC1=CC=CC=C1C(=O)O

Computed Nephrotoxicity Probability:
0.0130

Diagnostic Verdict: NON-NEPHROTOXIC
```

- Callable function: raw SMILES string → toxicity probability, end to end
- Recomputes descriptor + fingerprint matrix on the fly for any new molecule
- Reindexes features to the exact MI-selected 300-feature training schema

11 · LIMITATIONS

Stated Plainly, Not Hidden

Dataset scale

565 compounds is small for deep learning — the direct, expected explanation for DNN underperformance.

Dimensionality pressure

2175 raw features against 565 samples requires aggressive MI filtering, risking information loss.

Synthetic data dependency

SMOTE generates interpolated, not real, toxic-class training examples.

Chemical space coverage

Scaffold diversity is unconfirmed — the model's applicability domain outside this set is unverified.

Dataset is a published benchmark, not novel data

This 565-compound set matches Amin, Kar & Piotto (2026, Sci. Rep.) exactly — this work is a comparative model study on their data, not new data collection.

Scientific credibility means stating limitations as directly as results.

An End-to-End, Interpretable Nephrotoxicity Screen

SMILES Input → Feature Engineering → Model Tournament → Explainability → Live Screening

565
Compounds Screened

0.898
Best ROC-AUC (XGBoost)

Live
SMILES-to-Score Inference

- XGBoost identified as strongest model: 0.898 ROC-AUC, outperforming RF and a DNN
- SHAP explainability grounds predictions in interpretable structural attributions
- Live inference engine functionally demonstrated on an unseen compound
- Positioned as an early-stage triage layer — not a replacement for wet-lab toxicology assays